gabor filter

A **Gabor filter** is **a linear, bandpass filter used in image processing for texture analysis, feature extraction, and edge detection**. Its design is inspired by the functions of simple cells in the human visual system.

How It Works

A Gabor filter in the spatial domain is a Gaussian kernel function modulated by a sinusoidal plane wave. The kernel has real and imaginary components and can be adjusted with several parameters to tune its behavior:

- **Wavelength (λ):** Determines the frequency of the sinusoidal component, allowing the filter to detect specific frequency content in the image.
- **Orientation (θ):** Specifies the direction of the parallel stripes of the Gabor function, making the filter sensitive to structures at a particular angle.
- **Spatial Extent (σ, γ):** Controls the size and aspect ratio of the Gaussian envelope, which localizes the filter's effect in the image.

By creating a bank of filters with different wavelengths and orientations, an image can be analyzed for various directional and frequency information in a localized region, producing a comprehensive texture representation.

Primary Applications

Gabor filters are widely used in computer vision and pattern analysis due to their effectiveness in characterizing image features.

- **Biometrics:** A major application area, including **fingerprint recognition**, **iris recognition**, and **face recognition/detection**.

Ask anything

- **General Object Recognition:** Gabor features have been used to create robust object representations, sometimes in conjunction with modern methods like Convolutional Neural Networks (CNNs).

Would you like to know more about the **mathematical formula** or how to **implement** a Gabor filter in practice?

Say, I have 2 gabor filters with same frequency but one has bandwidth of 1 octave and second has bandwidth of 2 octav...

only responds to a very specific range of patterns that match its center frequency almost exactly.

Gabor Filters - Encyclopedia.pub

Dec 19, 2023 — The filter parameters, such as the wavelength, orientation, and spatial extent, can be adjusted to produce various filter properti...

Encyclopedia.pub

- **Spatial Characteristics:** It is a **large spatial filter** with many oscillations in its kernel. This means it "blurs" or averages features over a larger area to ensure it has correctly identified the specific frequency.

Gabor filter and its applications? - ResearchGate

Jul 9, 2018 — For document image processing, Gabor features are ideal for identifying the [illegible] tilingual document. 2. Gab...

ResearchGate

- **Best For:** Detecting subtle differences in texture density or precise repeating patterns (e.g., fine fabric weaves).

2 Octave Bandwidth Filter Show all

- **Artifacts Captured:** Sharp edges, isolated lines, and localized "shocks" in the image.
- **Frequency Specificity:** This filter is "wide" in the frequency domain. It will respond to a broader range of frequencies surrounding the center frequency.
- **Spatial Characteristics:** It is a **compact spatial filter** with very few oscillations. Because it covers a wider frequency band, it is much more "localized" in space.
- **Best For:** Edge detection and identifying the exact position of a feature, as it produces a sharper response at localized transitions.

Feature	1 Octave (Narrow)	2 Octaves (Wide)
Kernel Size	Large (more oscillations)	Small (fewer oscillations)
Frequency Response	Highly Selective	Broad
Spatial Localization	Poor (averages over a large area)	Excellent (tightly localized)
Image Feature	Smooth, repeating textures	Sharp edges and localized

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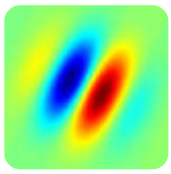
If you'd like, I can explain how to **adjust your filter parameters** (like sigma and lambda) to achieve these specific octave bandwidths in code.

22 sites

Understanding the Gabor filter function

Feb 28, 2014 — Tl;dr: bandwidths control the widths of the kernel. It is an inverse proportional relation. Eg. narrower bandwidth -> larger...

Signal Processing Stack Exchange



Gabor Feature - an overview | ScienceDirect Topics

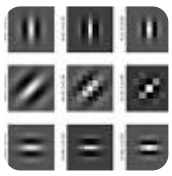
3. Applications of Gabor Features in Computer Vision and Pattern Recognition * Gabor features are particularly effective in extrac...

ScienceDirect.com

Gabor Filter - an overview | ScienceDirect Topics

Gabor filters [128–132] are a family of linear, frequency, and orientation-selective filters. They have been chosen for texture an...

ScienceDirect.com

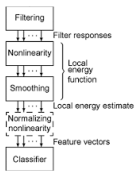


Show all

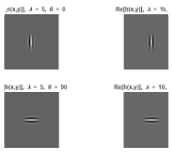
Filtering for Texture Classification: A Comparative Study

2.4. 3 Optimized Representation Gabor Filter Bank In order to tune the filters to the characteristics of the underlying textures, ...

IEEE Computer Society



gabor - Create Gabor filter or Gabor filter bank - MATLAB

Example: gabor(wavelength,orientation,"SpatialFrequencyBandwidth",2) creat...

MathWorks